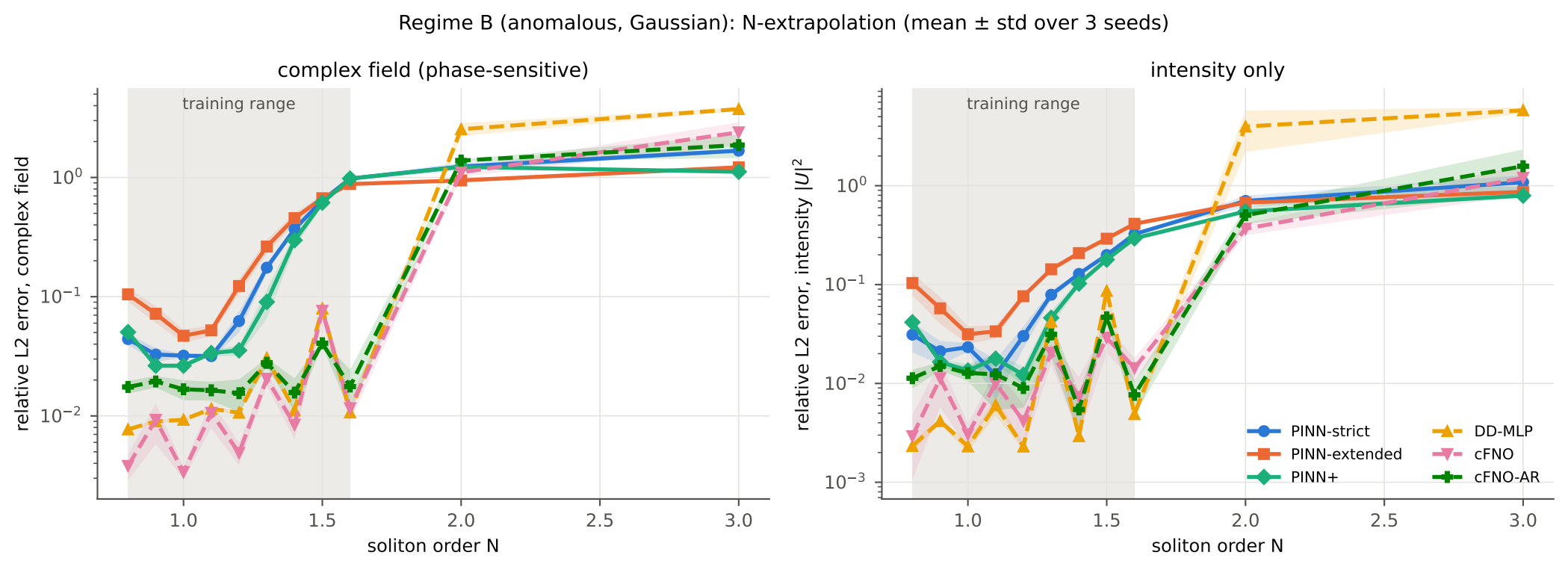
Regime B (anomalous, Gaussian): N-extrapolation (mean ± std over 3 seeds)
complex field (phase-sensitive)
intensity only
training range
training range
relative L2 error, complex field
relative L2 error, intensity $|U|^2$
10^{0}
10^{-1}
10^{-2}
10^{-3}
1.0
1.5
2.0
2.5
3.0
soliton order N
soliton order N
PINN-strict
PINN-extended
PINN+
DD-MLP
cFNO
cFNO-AR